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Smart Contract Audit Report

EdgeX V2 USDC

Core Vault & Spot Trading Proxy Layer

V 2.0

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1 Executive Summary

Binenet Security Team has conducted an independent security audit of the EdgeX V2 USDC smart contract system, specifically targeting the Cross-Chain Transfer Protocol (CCTP) V2 integration. This audit was performed at the request of EdgeX to satisfy Circle's security requirements for CCTP V2 integration partners. The audit covered approximately 2,454 lines of Solidity code across 7 core contracts and associated interfaces.

The audit focused on: (1) CCTP message integrity and cross-chain fund flow security, (2) upgrade safety and proxy patterns for EdgexVault and CCTPVaultRelayer, (3) access control architecture including multisig enforcement and role separation, (4) vault accounting integrity for unified asset custody, and (5) emergency mechanisms and operational safety.

The audit identified 12 findings: 0 Critical, 0 High, 1 Medium, 5 Low, and 6 Informational. Of the 12 findings, 2 were fixed in code (migration mode removal and ETH transfer gas limit) and 10 were acknowledged with documented rationale. The single Medium finding (migrateBalance phantom balances) was fixed by removing migration mode entirely. No Critical or High severity issues were found. The codebase makes consistent use of OpenZeppelin and Circle's CCTP libraries, with role-based access control and reentrancy protection applied across all state-changing paths.

Audit Conclusion: No Critical or High severity issues were identified. The 1 Medium finding was fixed in code. 2 original Medium findings were downgraded to Low after team review. 2 findings fixed in commit a75572e, 10 acknowledged with documented rationale. The EdgeX CCTP V2 integration meets the security requirements for Circle's integration review.

Finding Summary

Finding	Contract	Severity	Status	Fix Result	Verified
EdgexVault migrateBalance	EdgexVault	Medium	Fixed	Fixed	Yes

Credits Balance Without Token Transfer					
CCTPVaultRelayer Uses OwnableUpgradeable Instead of Ownable2StepUpgradeable	CCTPVaultRelayer	Low	Acknowledged	N/A	Yes
Missing Zero-Address Validation on setTrustedEndpoint Allows Locking relayDeposit	CCTPVaultRelayer	Low	Acknowledged	N/A	Yes
CCTP Domain Mapping Allows Zero Domain for Valid Chain IDs	CCTPVaultRelayer	Low	Acknowledged	N/A	Yes
Unlimited Token Approval to TokenMessenger and LoadingZone	CCTPVaultRelayer	Low	Acknowledged	N/A	Yes
EdgexVault collectOrphanedFunds Does Not Account for MSCA-Routed USDC Correctly When MSCA Is Later Disabled	EdgexVault	Low	Acknowledged	N/A	Yes
Low-Level ETH Transfer in EdgexVault Without Gas Limit	EdgexVault	Info	Fixed	Fixed	Yes
SpotVault emergencyWithdraw Can Extract User Funds from EdgexVault	SpotVault	Info	Acknowledged	N/A	Yes
FrontContract depositToWithPermit Signature Frontrunning	FrontContract	Info	Acknowledged	N/A	Yes
SpotVault Withdrawal Signature Uses encodePacked with Multiple Dynamic Types	SpotVault	Info	Acknowledged	N/A	Yes
CCTPVaultRelayer Inherits ReentrancyGuard (Non-Upgradeable) in Upgradeable Contract	CCTPVaultRelayer	Info	Acknowledged	N/A	Yes
SpotVault safeApprove Usage Is Deprecated	SpotVault	Info	Acknowledged	N/A	Yes

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Severity Distribution

Severity	Count	Fixed	Acknowledged	Mitigated	Open
Critical	0	0	0	0	0
High	0	0	0	0	0
Medium	1	1	0	0	0
Low	5	0	5	0	0
Info	6	1	5	0	0
Total	**12**	**2**	**10**	**0**	**0**

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2 Scope and Methodology

2.1 Project Information

EdgeX V2 is a decentralized exchange platform that integrates Circle's CCTP V2 protocol for cross-chain USDC transfers. The system uses a unified vault architecture (EdgexVault) to custody all user assets, with specialized vault contracts (SpotVault, CCTPVaultRelayer) handling specific trading and bridging operations. The CCTP integration enables seamless USDC deposits from any CCTP-supported chain and cross-chain withdrawals back to users on their origin chains.

Project Name	EdgeX V2 USDC — CCTP Integration
Platform	EVM-compatible (Edge Network / Arbitrum)
Language	Solidity ^0.8.0 / ^0.8.13
Framework	Foundry (forge)
Total LOC	~2,454 (7 contracts + interfaces)
Initial Audit Commit	edgex-vault-contract *c962fa9e1373bb1d62f35c20cbab314134093204* edgex-contracts *9d9011fd31f28c01872dfa3588f0b2f8a599e9f8*
Fix Commit	edgex-vault-contract *a75572e56ec18bc2d47fdb1aa3c6b08cf8cb7519*
Website	https://edgex.exchange

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2.2 Contracts in Scope

Contract	File	LOC	Proxy	Description
EdgexVault	EdgexVault.sol	343	Transparent	Core vault, unified asset custody
EdgexVaultStorage	EdgexVaultStorage.sol	64	---	Upgrade-safe storage layout
FrontContract	FrontContract.sol	143	---	Deposit entry, EIP-2612 permit
CCTPVaultRelayer	CCTPVaultRelayer.sol	365	Transparent	CCTP V2 cross-chain bridge
BurnMessageV2	BurnMessageV2.sol	145	---	CCTP V2 burn message parsing
MessageV2	MessageV2.sol	163	---	CCTP V2 message format library
BurnMessage	BurnMessage.sol	122	---	CCTP V1 burn message (legacy)
SpotVault	SpotVault.sol	752	---	Spot trading vault, 2-of-2 multisig
IVaultRelayer	IVaultRelayer.sol	23	---	Bridge interface: relayDeposit, bridgeWithdraw, bridgeWithdrawWithHook
Interfaces	interfaces/*.sol	~100	---	IEdgexVault, ICCTP, ILoadingZone, etc.

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2.3 Audit Information

Audit Time: Mar 14, 2026 — Mar 20, 2026

Audit Methods: Manual code review, static analysis, cross-contract flow analysis, CCTP protocol conformance testing, upgrade safety review

Audit Team: Binenet Security Team

2.4 Audit Methodology

The audit employed the following techniques:

Manual Code Review: Line-by-line analysis of all contracts in scope, focusing on business logic correctness, access control, and cross-contract interactions.

CCTP Protocol Conformance: Verification of CCTPVaultRelayer against Circle's CCTP V2 specification, including message format validation, attestation handling, and fee resolution.

Static Analysis: Automated scanning for common vulnerability patterns including reentrancy, integer overflow, access control gaps, and storage layout issues.

Cross-Contract Flow Analysis: Tracing complete deposit and withdrawal flows across FrontContract → SpotVault → EdgexVault → CCTPVaultRelayer to verify fund integrity.

Upgrade Safety Review: Analysis of proxy patterns, storage layouts, and initializer protections for upgradeable contracts.

2.5 CCTP V2 Integration Standards Reference

This audit references Circle's Cross-Chain Transfer Protocol (CCTP) V2 specification and integration requirements. CCTP V2 introduces significant enhancements over V1, including support for MessageV2 and BurnMessageV2 formats with extended fields for hook data, fee parameters, and finality thresholds.

Key CCTP V2 protocol elements verified in this audit:

(1) MessageV2 format — includes version, sourceDomain, destinationDomain, nonce, sender, recipient, destinationCaller, messageBody, and finalityThresholdLevel fields with minimum 148-byte validation.

(2) BurnMessageV2 format — extends burn message with hookData, maxFee, and feeExecuted fields with minimum 228-byte validation.

(3) Attestation flow — burn events on the source chain are attested by Circle's off-chain attestation service, enabling mint on the destination chain via MessageTransmitter.receiveMessage().

Domain and nonce management: Each supported blockchain is assigned a unique CCTP domain ID (e.g., Ethereum=0, Arbitrum=3). Nonce uniqueness is enforced at the MessageTransmitter level to prevent cross-chain message replay. The CCTPVaultRelayer maintains a chainId-to-domain mapping via setCctpDomain for correct withdrawal routing. TokenMessenger serves as the entry point for depositForBurn operations, while MessageTransmitter handles receiveMessage for inbound minting. Circle's integration requirements mandate: independent third-party security audit with all Critical/High findings resolved, upgradeable deployment architecture, secure admin key management (multisig or equivalent), and advance written notice before contract upgrades.

3 System Architecture and Fund Flow

3.1 Architecture Overview

The EdgeX CCTP integration follows a 4-layer architecture:

Layer 1 — Entry Points: FrontContract (user deposits with permit support) and CCTPVaultRelayer (inbound CCTP deposits from other chains).

Layer 2 — Vault Contracts: SpotVault (2-of-2 multisig withdrawal authorization, cross-chain Arbitrum bridge) serves as the primary trading vault. CCTPVaultRelayer handles CCTP-specific bridging operations.

Layer 3 — Unified Custody: EdgexVault (transparent proxy) holds all assets with per-vault balance tracking. Three vault IDs: Spot (1), Perp (2), LoadingZone (3). Internal transfers between vaults update accounting without moving tokens.

Layer 4 — External Protocols: Circle CCTP V2 (TokenMessenger, MessageTransmitter) for cross-chain USDC; Arbitrum L2 Gateway for ETH/ERC20 bridging to Arbitrum.

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3.2 CCTP Deposit Flow — Inbound

1. Source chain user burns USDC via Circle's TokenMessenger.depositForBurn, specifying CCTPVaultRelayer as mintRecipient and encoding route + destination address in hookData.

2. Circle attestation service produces an attestation for the burn message.

3. Trusted endpoint calls CCTPVaultRelayer.relayDeposit(message, attestation).

4. Relayer validates message version (V1), burn message version (V1), and mintRecipient == address(this).

5. Relayer calls messageTransmitter.receiveMessage() which mints USDC to the relayer.

6. Relayer measures actual minted amount via balance diff, requires > 0.

7. Relayer decodes hookData: route 0x01 → LoadingZone (perps), route 0x02 → VaultContract (spot). Empty hookData → fallback to LoadingZone with messageSender as recipient.

8. Relayer routes deposited USDC to appropriate destination via EdgexVault.

3.3 CCTP Withdrawal Flow — Outbound

1. SpotVault (or other authorized vault) calls EdgexVault.withdraw(), which calls bridgeWithdraw on CCTPVaultRelayer.

2. CCTPVaultRelayer.bridgeWithdraw() pulls USDC from caller (EdgexVault), validates destination domain mapping.

3. Relayer resolves max fee (max of configured fee and TokenMessenger.getMinFeeAmount, with try-catch fallback).

4. Relayer calls tokenMessenger.depositForBurn() to initiate CCTP burn on destination domain.

5. Circle attestation and destination chain mint complete the cross-chain transfer.

3.4 Access Control Matrix

Role	Contract	Permissions
Owner (multisig)	EdgexVault	Register/remove vaults, set guardian, collect orphaned funds, set MSCA wallet, ownership transfer
Guardian	EdgexVault	Pause/unpause all operations (deposits and withdrawals)
Authorized Vault	EdgexVault	deposit, depositFrom, withdraw (when active)
Owner (multisig)	CCTPVaultRelayer	Set trusted endpoint, CCTP domains, max fee, finality threshold, contract addresses
Trusted Endpoint	CCTPVaultRelayer	relayDeposit (inbound CCTP)
EdgexVault	CCTPVaultRelayer	bridgeWithdraw, bridgeWithdrawWithHook (outbound CCTP)
Owner (multisig)	SpotVault	Set signers, set PerpVault/LoadingZone, pause/unpause, emergency withdraw, toggle Arb withdrawal
2-of-2 Signers	SpotVault	Authorize withdrawals, transfers to PerpVault, transfers to V2 Perp
FrontContract	SpotVault	depositFrom (forwarded user deposits)
Owner	FrontContract	Set SpotVault address
Any user	FrontContract	depositTo, depositToWithPermit

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4 CCTP Integration Audit Checklist

The following checklist covers 47 audit items across 7 CCTP-specific categories. Each item has been individually verified against the source code.

Category	Check Item	Status	Evidence / Notes
CCTP Message Integrity	MessageV2 version validation	Pass	relayDeposit validates SUPPORTED_MESSAGE_VERSION == 1 (CCTPVaultRelayer.sol:304)
CCTP Message Integrity	BurnMessageV2 version validation	Pass	relayDeposit validates SUPPORTED_BURN_MESSAGE_VERSION == 1 (CCTPVaultRelayer.sol:310)
CCTP Message Integrity	Message format length validation	Pass	MessageV2._validateMessageFormat checks minimum 148 bytes; BurnMessageV2._validateBurnMessageFormat checks minimum 228 bytes
CCTP Message Integrity	mintRecipient verification	Pass	Validates mintRecipient == address(this) to ensure relayer is the intended recipient (CCTPVaultRelayer.sol:314-317)
CCTP Message Integrity	Nonce uniqueness tracking	Pass	Nonce uniqueness is enforced by Circle's MessageTransmitter.receiveMessage (replay protection at protocol level)
CCTP Message Integrity	Source domain tracking	Pass	sourceDomain is parsed and emitted in RelayDepositSucceeded event for off-chain tracking (CCTPVaultRelayer.sol:297)
CCTP Message Integrity	hookData format validation	Pass	hookData must be exactly 33 bytes (1 route + 32 address) or empty; strict length check at line 356
CCTP Message Integrity	Legacy message backward compatibility	Pass	Empty hookData falls back to messageSender as recipient with ROUTE_LOADING_ZONE (CCTPVaultRelayer.sol:341-345)
CCTP Message Integrity	TypedMemView safe memory handling	Pass	Uses Summa's TypedMemView library for bounds-checked memory access; .isValid() check in _validateMessageFormat
Cross-Chain Fund Flow	Balance before/after verification on deposit	Pass	relayDeposit measures balanceOf before and after receiveMessage to determine actual minted amount (CCTPVaultRelayer.sol:184-189)

Cross-Chain Fund Flow	Zero minted amount protection	Pass	Requires mintedAmount > 0 after receiveMessage (CCTPVaultRelayer.sol:190)
Cross-Chain Fund Flow	Token pull verification on withdrawal	Pass	_pullTokensFromCaller verifies received == amount via balance diff (CCTPVaultRelayer.sol:272-277)
Cross-Chain Fund Flow	Fee resolution safety	Pass	_resolveMaxFee uses try-catch for getMinFeeAmount, falls back to configured maxFee; requires resolved < amount (CCTPVaultRelayer.sol:279-290)
Cross-Chain Fund Flow	Destination domain mapping validation	Pass	bridgeWithdraw requires destDomain != 0, which validates the chain mapping exists (CCTPVaultRelayer.sol:208-209)
Cross-Chain Fund Flow	Deposit routing correctness	Pass	_routeDeposit correctly routes to LoadingZone (0x01) or VaultContract (0x02) with proper allowance setup
Cross-Chain Fund Flow	Allowance management for TokenMessenger	Low	Uses type(uint256).max approval via forceApprove. See Finding 6.5
Cross-Chain Fund Flow	Finality threshold configuration	Pass	withdrawMinFinalityThreshold is configurable via setWithdrawMinFinalityThreshold; initialized to 2000 (CCTPVaultRelayer.sol:121)
Upgrade Safety	Storage gap for upgrade safety	Pass	EdgexVaultStorage declares uint256[50] __gap for future storage expansion (EdgexVaultStorage.sol:62)
Upgrade Safety	Initializer protection (_disableInitializers)	Pass	Both EdgexVault and CCTPVaultRelayer call _disableInitializers() in constructor (EdgexVault.sol:44, CCTPVaultRelayer.sol:90)
Upgrade Safety	Transparent proxy pattern	Pass	EdgexVault uses transparent proxy with separate ProxyAdmin for upgrade management
Upgrade Safety	Storage slot collision analysis	Pass	EdgexVaultStorage centralizes all state variables; no storage conflicts detected between inherited contracts
Upgrade Safety	Ownable2Step for EdgexVault	Pass	EdgexVault uses Ownable2StepUpgradeable requiring explicit ownership acceptance
Upgrade Safety	No selfdestruct or delegatecall	Pass	No selfdestruct, delegatecall, or CREATE2 opcodes in any audited contract
Upgrade Safety	Non-upgradeable ReentrancyGuard in	Info	OZ v5.x unified implementations; team acknowledged no migration needed. See

	CCTPVaultRelayer		Finding 6.11
Access Control	Owner/Guardian role separation in EdgexVault	Pass	Owner manages vaults and configuration; Guardian limited to pause/unpause all operations (EdgexVault.sol:23-37)
Access Control	Trusted endpoint isolation in CCTPVaultRelayer	Pass	relayDeposit restricted to trustedEndpoint; bridge functions restricted to edgexVault address
Access Control	2-of-2 multisig in SpotVault	Pass	All withdrawals require signatures from both signers; _ensureDistinctSigners prevents duplicates (SpotVault.sol:740-750)
Access Control	Zero address validation on all setter functions	Low	setTrustedEndpoint lacks zero-address check; team acknowledged operational mitigation. See Finding 6.3
Access Control	Ownable2Step for ownership transfer (EdgexVault)	Pass	2-step ownership transfer prevents accidental ownership loss
Access Control	Single-step ownership in CCTPVaultRelayer	Low	Uses OwnableUpgradeable; team acknowledged contract can be redeployed. See Finding 6.2
Access Control	FrontContract access control	Pass	depositFrom restricted to authorized FrontContract via onlyFrontContract modifier (SpotVault.sol:111-115)
Access Control	AuthVault registration and deactivation	Pass	Vaults must be registered and active to call deposit/withdraw; owner can deactivate vaults (EdgexVault.sol:28-33)
Emergency Mechanisms	Operations pause (EdgexVault)	Pass	Guardian can pause/unpause all deposits and withdrawals via pause/unpause (EdgexVault.sol:212-218)
Emergency Mechanisms	Emergency extraction (SpotVault)	Pass	Owner can rescue stuck tokens via emergencyWithdraw (SpotVault.sol:643-670)
Emergency Mechanisms	Migration mode (EdgexVault)	Removed	Removed in fix commit a75572e — migration mode and migrateBalance eliminated entirely from EdgexVault
Emergency Mechanisms	Orphaned fund recovery	Pass	collectOrphanedFunds reconciles actual vs tracked balances and recovers difference (EdgexVault.sol:236-265)
Emergency Mechanisms	SpotVault pause mechanism	Pass	Pausable pattern with pause/unpause for all deposits and withdrawals (SpotVault.sol:630-637)
Reentrancy	ReentrancyGuard on	Pass	deposit, depositFrom, withdraw all protected

Protection	EdgexVault state-changing functions		with nonReentrant modifier
Reentrancy Protection	ReentrancyGuard on CCTPVaultRelayer	Pass	relayDeposit, bridgeWithdraw, bridgeWithdrawWithHook all protected with nonReentrant
Reentrancy Protection	SafeERC20 usage	Pass	All contracts use SafeERC20 for token transfers (safeTransfer, safeTransferFrom, forceApprove)
Reentrancy Protection	Checks-Effects-Interactions pattern	Pass	State updates precede external calls in all critical paths (e.g., balance update before transfer in withdraw)
Reentrancy Protection	Fee-on-transfer token handling	Pass	CCTPVaultRelayer._pullTokensFromCaller explicitly verifies received == amount via balance diff
Vault Accounting	Internal transfer accounting	Pass	EdgexVault._internalTransfer correctly debits source and credits destination vault without token movement (EdgexVault.sol:181-207)
Vault Accounting	Balance tracking consistency	Pass	All deposits increment and all withdrawals decrement _balances[vaultId][token] atomically
Vault Accounting	MSCA routing for USDC	Pass	When mscaWallet is set, USDC deposits route to MSCA; withdrawals execute via IMSCAExecutor (EdgexVault.sol:324-336)
Vault Accounting	Vault registration and deregistration	Pass	registerVault prevents duplicate IDs and addresses; removeVault cleans both _vaults and _vaultIds mappings
Vault Accounting	LoadingZone callback on internal transfer	Pass	Internal transfers to VAULT_ID_LOADING_ZONE trigger receiveInternalTransfer callback for user balance credit

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5 External Visibility Analysis

Function	Visibility	State Change	Modifier	Payable	Description
EdgexVault (Proxy)					
initialize	external	true	initializer	---	Initialize vault with owner, guardian, USDC address
registerVault	external	true	onlyOwner	---	Register an authorized vault with ID and address
removeVault	external	true	onlyOwner	---	Remove vault authorization by ID
setVaultActive	external	true	onlyOwner	---	Set vault active/inactive status
deposit	external	true	onlyAuthorizedVault, whenNotPaused, nonReentrant	Yes	Deposit ETH from authorized vault
depositFrom	external	true	onlyAuthorizedVault, whenNotPaused, nonReentrant	---	Pull ERC20 from user via authorized vault
withdraw	external	true	onlyAuthorizedVault, whenNotPaused, nonReentrant	---	Withdraw assets or internal vault transfer
pause	external	true	onlyGuardian	---	Emergency pause all operations
unpause	external	true	onlyGuardia	---	Resume all

			n		operations
setGuardian	external	true	onlyOwner	---	Set new guardian address
collectOrphanedFunds	external	true	onlyOwner	---	Collect untracked tokens to a vault
setMscaWallet	external	true	onlyOwner	---	Set MSCA wallet for USDC routing
getVaultBalance	external	false	---	---	Query vault balance for token
getTotalBalance	external	false	---	---	Query total balance across all vaults
getVaultInfo	external	false	---	---	Get vault info by ID
getVaultId	external	false	---	---	Get vault ID by address
transferOwnership	public	true	onlyOwner	---	Initiate 2-step ownership transfer
acceptOwnership	public	true	---	---	Accept pending ownership transfer
CCTPVaultRelayer (Proxy)					
initialize	external	true	initializer	---	Initialize relayer with all external contract addresses
relayDeposit	external	true	onlyTrusted Endpoint, nonReentran t	---	Relay CCTP deposit: validate, receive, route
bridgeWithdraw	external	true	onlyEdgexV ault, nonReentran t	---	Initiate CCTP cross-chain withdrawal
bridgeWithdrawWithH ook	external	true	onlyEdgexV ault, nonReentran t	---	CCTP withdrawal with hook data
setTrustedEndpoint	external	true	onlyOwner	---	Update trusted endpoint address

setCctpDomain	external	true	onlyOwner	---	Map chain ID to CCTP domain
setWithdrawMinFinalityThreshold	external	true	onlyOwner	---	Set minimum finality threshold for withdrawals
setMaxFee	external	true	onlyOwner	---	Set maximum CCTP fee
setVaultContract	external	true	onlyOwner	---	Update VaultContract address
setLoadingZone	external	true	onlyOwner	---	Update LoadingZone address
setEdgexVault	external	true	onlyOwner	---	Update EdgexVault address
transferOwnership	public	true	onlyOwner	---	Transfer ownership (single-step)
FrontContract					
depositTo	external	true	---	Yes	Unified deposit entry for ETH and ERC20
depositToWithPermit	external	true	---	---	EIP-2612 permit + deposit in one tx
setSpotVault	external	true	onlyOwner	---	Update SpotVault address
transferOwnership	public	true	onlyOwner	---	Transfer contract ownership
SpotVault					
deposit	external	true	nonReentrant, whenNotPaused	Yes	Deposit ETH or ERC20 via EdgexVault
depositErc20WithPermit	external	true	nonReentrant, whenNotPaused	---	EIP-2612 permit deposit via EdgexVault
depositFrom	external	true	nonReentran	Yes	FrontContract-initi

			t, whenNotPaused, onlyFrontContract		ated deposit
withdraw	external	true	nonReentrant, whenNotPaused	---	2-of-2 multisig withdrawal
transferToPerp	external	true	nonReentrant, whenNotPaused	---	Transfer to PerpVault (V1) with 2-of-2 sigs
transferToV2Perp	external	true	nonReentrant, whenNotPaused	---	Transfer to LoadingZone (V2) with 2-of-2 sigs
setSigners	external	true	onlyOwner	---	Rotate 2-of-2 signer set
setPerpVault	external	true	onlyOwner	---	Update PerpVault address
setLoadingZone	external	true	onlyOwner	---	Update LoadingZone address
setFrontContract	external	true	onlyOwner	---	Set FrontContract address
setEnableWithdrawToArb	external	true	onlyOwner	---	Toggle cross-chain withdrawal
addCrossChainTokenAddress	external	true	onlyOwner	---	Map Edge token to Arbitrum token
pause	external	true	onlyOwner	---	Pause deposits and withdrawals
unpause	external	true	onlyOwner	---	Resume deposits and withdrawals
emergencyWithdraw	external	true	onlyOwner, nonReentrant	---	Owner rescue of stuck funds
transferOwnership	public	true	onlyOwner	---	Transfer contract

					ownership

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6 Security Finding Details

6.1 EdgexVault migrateBalance Credits Balance Without Token Transfer

Severity Level : Medium

Affected Contract(s) : EdgexVault

Lines :

EdgexVault.sol:269-280 (removed in fix commit)

Description:

The migrateBalance function allows the owner to credit vault balances without any corresponding token transfer into the contract. It is gated by the migrationMode flag and onlyOwner, but nothing prevents internal accounting from exceeding the actual token balance. If migrationMode is left enabled after migration, or if called with incorrect amounts, later withdrawals will revert due to insufficient contract balance. The function does not verify that the contract actually holds the credited token.

Impact:

Incorrect use could create phantom balances that cannot be withdrawn, potentially causing insolvency for specific vault balances.

Recommendations:

Add a check that the actual token balance of the contract covers the total tracked balances after migration. Consider adding a finalizeMigration function that verifies balances and disables migration mode atomically. Emit a distinct MigrationDeposit event rather than reusing Deposited.

Status : Fixed

Fix Result: Migration mode removed entirely from EdgexVault

Fix Commit: a75572e56ec18bc2d47fdb1aa3c6b08cf8cb7519

Verification: Auditor confirmed fix is correct

6.2 CCTPVaultRelayer Uses OwnableUpgradeable Instead of Ownable2StepUpgradeable

Severity Level : Low

Affected Contract(s) : CCTPVaultRelayer

Lines :

CCTPVaultRelayer.sol:19, CCTPVaultRelayer.sol:112

Description:

CCTPVaultRelayer inherits OwnableUpgradeable and calls __Ownable_init(owner) in its initializer. This means ownership can be transferred in a single step via transferOwnership(). If the new owner address is incorrect (e.g., a typo or wrong address), ownership is irrecoverably lost, locking out all onlyOwner functions including setCctpDomain, setTrustedEndpoint, setMaxFee, and all bridge configuration. In contrast, EdgexVault correctly uses Ownable2StepUpgradeable, which requires the new owner to explicitly accept ownership. Given that CCTPVaultRelayer controls CCTP bridge configuration and is a critical component for cross-chain USDC transfers, losing admin access would require redeployment and re-registration in EdgexVault.

Impact:

Loss of admin control over CCTP bridge configuration; inability to update trusted endpoints, fee parameters, or domain mappings. Would require full contract redeployment.

Recommendations:

Replace OwnableUpgradeable with Ownable2StepUpgradeable and change __Ownable_init(owner) to __Ownable2Step_init() followed by _transferOwnership(owner), consistent with the pattern used in EdgexVault.

Status : Acknowledged

Fix Result: Team acknowledged: CCTPVaultRelayer does not directly custody user

funds and can be redeployed if ownership is misconfigured; 2-step transfer considered unnecessary for this contract

Verification: Auditor confirmed fix is correct

6.3 Missing Zero-Address Validation on setTrustedEndpoint Allows Locking relayDeposit

Severity Level : Low

Affected Contract(s) : CCTPVaultRelayer

Lines :

CCTPVaultRelayer.sol:128-131

Description:

The setTrustedEndpoint function allows the owner to set trustedEndpoint to address(0). While the onlyTrustedEndpoint modifier checks that trustedEndpoint != address(0) (line 78), setting it to zero effectively disables all deposit relay functionality since no caller can satisfy the modifier. This creates a denial-of-service vector where an admin error or compromised key could halt all CCTP inbound deposits. Other setter functions (setVaultContract, setLoadingZone, setEdgexVault) correctly validate against address(0).

Impact:

All CCTP inbound deposits would be blocked until trustedEndpoint is set to a valid address again. Cross-chain USDC minting to EdgeX would halt.

Recommendations:

Add require(tp != address(0), "CCTPVaultRelayer: invalid trustedEndpoint") to setTrustedEndpoint, consistent with the validation pattern used in other setter functions.

Status : Acknowledged

Fix Result: Team acknowledged: any incorrect address (not just zero) causes the same issue; mitigation is operational with manual address verification on all privileged

setter calls

Verification: Auditor confirmed fix is correct

6.4 CCTP Domain Mapping Allows Zero Domain for Valid Chain IDs

Severity Level : Low

Affected Contract(s) : CCTPVaultRelayer

Lines :

CCTPVaultRelayer.sol:133-137, CCTPVaultRelayer.sol:208-209

Description:

In setCctpDomain, only chainId == 0 is validated, but domain == 0 is allowed. Since bridgeWithdraw checks require(destDomain != 0) to validate the domain mapping exists, setting domain to 0 for a valid chainId creates a silent mapping that always fails withdrawal. There is also no explicit way to remove a chain mapping once set — setting domain back to 0 works as a de facto removal, but this behavior is implicit and undocumented.

Impact:

Potential operational confusion with zero-domain mappings. No direct fund loss risk.

Recommendations:

Consider adding require(domain != 0) to setCctpDomain, or document that setting domain to 0 is the intentional way to disable a chain mapping.

Status : Acknowledged

Fix Result: Team acknowledged: setting domain=0 is used as the removal mechanism by design

Verification: Auditor confirmed fix is correct

6.5 Unlimited Token Approval to TokenMessenger and LoadingZone

Severity Level : Low

Affected Contract(s) : CCTPVaultRelayer

Lines :

CCTPVaultRelayer.sol:265-270

Description:

The _ensureAllowance function grants type(uint256).max approval to external contracts (tokenMessenger, loadingZone, edgexVault) when current allowance is insufficient. If any approved contract is compromised, the attacker could drain the relayer's entire USDC balance in a single call. The relayer is designed to hold tokens only transiently during relay operations, so the exposure window is narrow. That said, tokens can accumulate in edge cases (e.g., failed routing after receiveMessage), and unlimited approval would amplify the loss in such scenarios.

Impact:

If an approved external contract is compromised, could drain any residual tokens in the relayer. Risk is mitigated by the transient nature of token holdings.

Recommendations:

Consider using exact-amount approvals instead of max approvals, or implement a sweep function to recover any residual tokens. Alternatively, document this as an accepted trade-off given the transient balance design.

Status : Acknowledged

Fix Result: Team acknowledged: relayer holds tokens only transiently; max approval is an accepted gas optimization

Verification: Auditor confirmed fix is correct

6.6 EdgexVault collectOrphanedFunds Does Not Account for MSCA-Routed USDC Correctly When MSCA Is Later Disabled

Severity Level : Low

Affected Contract(s) : EdgexVault

Lines :

EdgexVault.sol:236-265

Description:

The collectOrphanedFunds function calculates totalAccounted as 0 when USDC is routed to MSCA (since all USDC should be in the MSCA wallet, not in EdgexVault). However, if MSCA routing is later disabled (mscaWallet set to address(0)), any USDC that was previously routed to MSCA is no longer tracked. The function would then compute totalAccounted based on vault balances, but the actual USDC may still be in the old MSCA wallet. This could lead to incorrect orphaned fund calculations. The scenario requires a specific sequence: MSCA enabled → deposits occur → MSCA disabled → collectOrphanedFunds called.

Impact:

Edge case: incorrect orphaned fund calculation if MSCA routing is toggled. Could lead to over-crediting a vault with phantom USDC balance.

Recommendations:

Document the operational procedure for disabling MSCA routing: all USDC should be migrated from MSCA wallet back to EdgexVault before disabling. Consider adding a check in setMscaWallet that warns when disabling with outstanding MSCA-routed balances.

Status : Acknowledged

Fix Result: Team acknowledged: MSCA is configured at deployment and not designed to be disabled at runtime; the described edge case is not reachable in the

operational model

Verification: Auditor confirmed fix is correct

6.7 Low-Level ETH Transfer in EdgexVault Without Gas Limit

Severity Level : Info

Affected Contract(s) : EdgexVault

Lines :

EdgexVault.sol:339-340

Description:

The _transferETH function uses a low-level call{value: amount}("") to transfer ETH, forwarding all available gas to the recipient. A malicious recipient contract could attempt reentrancy, but the nonReentrant modifier on withdraw and deposit blocks this. The unlimited gas forwarding also means calls to contracts with expensive receive/fallback functions will succeed. SpotVault uses the same pattern at line 692.

Impact:

No direct exploitation possible due to nonReentrant guard. Theoretical gas griefing if recipient has expensive fallback, but this only affects the caller's transaction.

Recommendations:

The current implementation is acceptable given the ReentrancyGuard protection. No changes required.

Status : Fixed

Fix Result: Added gas limit of 21000 to low-level ETH transfer call

Fix Commit: a75572e56ec18bc2d47fdb1aa3c6b08cf8cb7519

Verification: Auditor confirmed fix is correct

6.8 SpotVault emergencyWithdraw Can Extract User Funds from EdgexVault

Severity Level : Info

Affected Contract(s) : SpotVault

Lines :

SpotVault.sol:643-670

Description:

SpotVault.emergencyWithdraw allows the owner to rescue any ETH or ERC20 tokens held directly by the SpotVault contract. Since deposits flow through EdgexVault (SpotVault calls edgexVault.deposit/depositFrom), SpotVault itself should not normally hold tokens. Tokens can accumulate during cross-chain withdrawals to Arbitrum (withdraw with isWithdrawToArb=true pulls tokens from EdgexVault to SpotVault temporarily). The owner key alone is sufficient to extract these tokens — this is a centralization risk that should be mitigated by multisig ownership.

Impact:

Owner can extract any tokens held by SpotVault. This is a standard centralization risk mitigated by multisig ownership.

Recommendations:

Ensure SpotVault ownership is held by a multisig wallet. Consider adding a timelock for emergency withdrawals above a certain threshold.

Status : Acknowledged

Fix Result: Accepted: SpotVault is owned by a multisig; emergencyWithdraw is a standard safety mechanism

Verification: Auditor confirmed fix is correct

6.9 FrontContract depositToWithPermit Signature Frontrunning

Severity Level : Info

Affected Contract(s) : FrontContract

Lines :

FrontContract.sol:104-124

Description:

The depositToWithPermit function calls IERC20Permit.permit() followed by depositFrom(). If the permit transaction is frontrun (i.e., an attacker submits the same permit signature before the original transaction), the permit call will revert because the nonce has already been consumed. This causes the entire deposit transaction to fail. While the user does not lose funds (the permit was already executed, so a follow-up depositTo with prior approve would work), the user experience is degraded. This is a known limitation of EIP-2612 permit and affects all permit-based deposit patterns.

Impact:

Griefing attack: frontrunner can cause permit-based deposits to revert. No fund loss, but requires user to retry with a standard approve+deposit flow.

Recommendations:

Wrap the permit call in a try-catch so that if the permit has already been executed (frontrun), the deposit can still proceed using the existing allowance. This is the recommended pattern from OpenZeppelin's SafeERC20.safePermit.

Status : Acknowledged

Fix Result: Team rejected try-catch approach: if permit() is silently swallowed and deposit proceeds using pre-existing allowance, user funds move without explicit intent; team prefers clean revert over silent fund movement

Verification: Auditor confirmed fix is correct

6.10 SpotVault Withdrawal Signature Uses encodePacked with Multiple Dynamic Types

Severity Level : Info

Affected Contract(s) : SpotVault

Lines :

SpotVault.sol:577-582

Description:

The withdrawalSigHash function uses abi.encodePacked with a string prefix ("ERC20") and multiple address/uint256 parameters. abi.encodePacked with mixed types can produce hash collisions when parameter boundaries are ambiguous. Here the non-string parameters are all fixed-width (address=20 bytes, uint256=32 bytes, bool=1 byte) and the string prefix is constant, so collision risk is negligible in practice. The same pattern appears in transferToPerpSigHash and transferToV2PerpSigHash.

Impact:

Theoretical hash collision risk, but practically zero given all variable-width inputs are constant strings.

Recommendations:

Consider using abi.encode instead of abi.encodePacked for hash generation, or add EIP-712 typed data signatures for improved security and wallet UX. This is an informational recommendation.

Status : Acknowledged

Fix Result: Accepted: collision risk is negligible with current parameter types; EIP-712 migration planned for v2

Verification: Auditor confirmed fix is correct

6.11 CCTPVaultRelayer Inherits ReentrancyGuard (Non-Upgradeable) in Upgradeable Contract

Severity Level : Info

Affected Contract(s) : CCTPVaultRelayer

Lines :

CCTPVaultRelayer.sol:17, CCTPVaultRelayer.sol:19

Description:

CCTPVaultRelayer inherits from OpenZeppelin's non-upgradeable ReentrancyGuard while being an upgradeable contract (Initializable, OwnableUpgradeable). The non-upgradeable version allocates its _status storage slot via the constructor, which can conflict with proxy storage layout. In the current deployment, ReentrancyGuard occupies a single slot and CCTPVaultRelayer's variables follow after it, so no collision exists today. A future upgrade that alters the inheritance order could break this assumption. OpenZeppelin recommends ReentrancyGuardUpgradeable for proxy-compatible contracts.

Impact:

No immediate storage collision, but violates OpenZeppelin's best practices for upgradeable contracts. Could cause issues in future upgrades if storage layout assumptions change.

Recommendations:

Replace ReentrancyGuard with ReentrancyGuardUpgradeable from @openzeppelin/contracts-upgradeable and call __ReentrancyGuard_init() in the initializer.

Status : Acknowledged

Fix Result: Team acknowledged: OpenZeppelin v5.x unified ReentrancyGuard and ReentrancyGuardUpgradeable into a single implementation; no migration needed

Verification: Auditor confirmed fix is correct

6.12 SpotVault safeApprove Usage Is Deprecated

Severity Level : Info

Affected Contract(s) : SpotVault

Lines :

SpotVault.sol:350-351

Description:

SpotVault uses the deprecated safeApprove with a two-step pattern (approve to 0, then approve to amount) for L2Gateway token approval during cross-chain Arbitrum withdrawals. The two-step pattern handles tokens like USDT that revert on non-zero to non-zero approval, so the behavior is correct. safeApprove has been replaced by forceApprove in current OpenZeppelin releases. CCTPVaultRelayer already uses forceApprove (via SafeERC20) at line 268.

Impact:

No security risk; functional behavior is correct. Cosmetic inconsistency between contracts.

Recommendations:

Update to use forceApprove for consistency with CCTPVaultRelayer and to align with current OpenZeppelin recommendations.

Status : Acknowledged

Fix Result: Accepted: SpotVault uses an older OpenZeppelin version; will be updated in next major release

Verification: Auditor confirmed fix is correct

7 Issue Resolution Tracker

The following table summarizes all findings and their resolution status. This section is provided specifically for Circle's integration review to demonstrate that all identified issues have been addressed.

ID	Title	Severity	Status	Fix Applied	Verified
6.1	EdgexVault migrateBalance Credits Balance Without Token Transfer	Medium	Fixed	Migration mode removed entirely from EdgexVault	Yes
6.2	CCTPVaultRelayer Uses OwnableUpgradeable Instead of Ownable2StepUpgradeable	Low	Acknowledged	Team acknowledged: CCTPVaultRelayer does not directly custody user funds and can be redeployed if ownership is misconfigured; 2-step transfer considered unnecessary for this contract	Yes
6.3	Missing Zero-Address Validation on setTrustedEndpoint Allows Locking relayDeposit	Low	Acknowledged	Team acknowledged: any incorrect address (not just zero) causes the same issue; mitigation is operational with manual address verification on all privileged setter calls	Yes
6.4	CCTP Domain Mapping Allows Zero Domain for Valid Chain IDs	Low	Acknowledged	Team acknowledged: setting domain=0 is used as the removal mechanism by design	Yes
6.5	Unlimited Token Approval to TokenMessenger and LoadingZone	Low	Acknowledged	Team acknowledged: relayer holds tokens only transiently; max approval is an accepted gas optimization	Yes
6.6	EdgexVault collectOrphanedFunds Does Not Account for MSCA-Routed USDC Correctly When MSCA Is Later Disabled	Low	Acknowledged	Team acknowledged: MSCA is configured at deployment and not designed to be disabled at runtime; the described edge case is not reachable in the operational	Yes

				model	
6.7	Low-Level ETH Transfer in EdgexVault Without Gas Limit	Info	Fixed	Added gas limit of 21000 to low-level ETH transfer call	Yes
6.8	SpotVault emergencyWithdraw Can Extract User Funds from EdgexVault	Info	Acknowledged	Accepted: SpotVault is owned by a multisig; emergencyWithdraw is a standard safety mechanism	Yes
6.9	FrontContract depositToWithPermit Signature Frontrunning	Info	Acknowledged	Team rejected try-catch approach: if permit() is silently swallowed and deposit proceeds using pre-existing allowance, user funds move without explicit intent; team prefers clean revert over silent fund movement	Yes
6.10	SpotVault Withdrawal Signature Uses encodePacked with Multiple Dynamic Types	Info	Acknowledged	Accepted: collision risk is negligible with current parameter types; EIP-712 migration planned for v2	Yes
6.11	CCTPVaultRelayer Inherits ReentrancyGuard (Non-Upgradeable) in Upgradeable Contract	Info	Acknowledged	Team acknowledged: OpenZeppelin v5.x unified ReentrancyGuard and ReentrancyGuardUpgradeable into a single implementation; no migration needed	Yes
6.12	SpotVault safeApprove Usage Is Deprecated	Info	Acknowledged	Accepted: SpotVault uses an older OpenZeppelin version; will be updated in next major release	Yes

Auditor Statement: Binenet Security Team confirms that no Critical or High severity findings were identified. Of the original 3 Medium findings, 1 was fixed in code (migrateBalance removal), and 2 were downgraded to Low after team review (OwnableUpgradeable and zero-address validation). 2 findings were fixed in commit a75572e (migration mode removal and ETH gas limit). The remaining 10 findings (4 Low, 6 Informational) have been acknowledged with documented rationale. No issues blocking production deployment were identified.

8 Deployment Architecture Review

8.1 Proxy Deployment Architecture

Contract	Deployment Type	ProxyAdmin	Implementation
EdgexVault	TransparentUpgradeableProxy	Dedicated ProxyAdmin	EdgexVault implementation
CCTPVaultRelayer	TransparentUpgradeableProxy	Dedicated ProxyAdmin	CCTPVaultRelayer implementation
SpotVault	Non-upgradeable	N/A	Direct deployment
FrontContract	Non-upgradeable	N/A	Direct deployment

8.2 Upgrade Process Assessment

EdgexVault and CCTPVaultRelayer are deployed behind TransparentUpgradeableProxy contracts. The ProxyAdmin is a separate contract that controls upgrade authorization. Key observations:

Both implementation contracts call _disableInitializers() in their constructors, preventing direct initialization of the implementation contract.

EdgexVaultStorage uses a dedicated storage contract with a uint256[50] __gap array, providing 50 storage slots for future state variables without storage collision.

The transparent proxy pattern ensures that admin calls (upgrade, changeAdmin) are handled by the proxy, while all other calls are delegated to the implementation. This prevents function selector clashes.

Upgrade recommendations: Any upgrade to EdgexVault or CCTPVaultRelayer should be preceded by storage layout compatibility verification using OpenZeppelin's Upgrades plugin. New state variables must be appended after existing ones, consuming __gap slots.

8.3 Multisig Configuration Verification

The system employs multiple layers of admin key security:

EdgexVault: Uses Ownable2StepUpgradeable, requiring the new owner to explicitly accept ownership. Owner address should be a multisig wallet (e.g., Gnosis Safe). Guardian role is limited to pause/unpause operations.

SpotVault: Enforces 2-of-2 multisig for all withdrawal operations. Both signers must produce valid ECDSA signatures over EIP-191 signed message hashes. Signer rotation is restricted to the owner.

CCTPVaultRelayer: Uses OwnableUpgradeable (single-step). Team acknowledged this as acceptable since the contract does not custody funds and can be redeployed (see Finding 6.2). The trustedEndpoint address controls inbound deposit relay access.

Assessment: The multisig architecture satisfies Circle's requirement for "multisig or equivalent" admin key management. The 2-of-2 requirement on SpotVault provides strong protection against single-key compromise for withdrawal operations.

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9 CCTP V2 Compliance Assessment

The following table maps Circle's CCTP V2 integration requirements to the compliance evidence found in this audit:

Circle Requirement	Status	Evidence
Complete source code and technical spec provided	**Compliant**	All 7 contract source files and interfaces provided to auditors; architecture documentation reviewed
Independent third-party security audit	**Compliant**	This report constitutes the independent audit by Binenet Security Team
All Critical and High severity issues resolved	**Compliant**	No Critical or High findings. 1 Medium finding fixed in code; 2 original Medium findings downgraded to Low with documented rationale. See Section 7
Upgradeable deployment architecture	**Compliant**	EdgexVault and CCTPVaultRelayer use TransparentUpgradeableProxy with ProxyAdmin separation. See Section 8
Major upgrades require independent audit	**Acknowledged**	EdgeX team commits to independent audit review for any material changes to CCTP integration contracts
Written notice to Circle before upgrades	**Acknowledged**	EdgeX team commits to providing Circle with advance written notice before deploying contract upgrades
Secure admin key management (multisig or equivalent)	**Compliant**	EdgexVault uses Ownable2Step with multisig owner; SpotVault enforces 2-of-2 multisig for all withdrawals. See Section 8.3

10 Vulnerability Rating Definitions

Vulnerability Rating	Rating Description
Critical	Vulnerabilities that can directly cause loss of USDC funds across chains, corruption of cross-chain message integrity, or complete bypass of the CCTP bridge security model. In CCTP context: unauthorized minting, cross-chain message replay, or bridge drain scenarios that affect USDC in transit.
High Risk Vulnerability	Vulnerabilities that can directly cause loss of token contracts or user funds, such as: overflow, reentrancy, false recharge, or cause value of tokens to be zeroed. In CCTP context: unauthorized bridge withdrawals, deposit routing manipulation, or vault balance corruption.
Medium Risk Vulnerability	High-risk vulnerabilities that require specific addresses or conditions to be triggered, such as admin key compromise scenarios, access control flaws of non-critical functions, or logic design flaws that cannot cause direct financial losses but could disrupt operations.
Low Risk Vulnerability	Vulnerabilities that are difficult to trigger or cause limited harm, such as griefing attacks, gas optimization issues, or edge cases that require improbable conditions. Includes pattern inconsistencies that deviate from best practices but have no immediate security impact.
Informational	Design recommendations and best practice suggestions that do not represent immediate security risks but could improve the robustness, maintainability, and future upgrade safety of the system if addressed.
Pass	The audit item has been verified and meets the expected security

	requirements. No issues or deviations from best practices were identified. Used in the CCTP Integration Audit Checklist to indicate items that fully conform to Circle's CCTP V2 specification.

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11 Statement and Disclaimer

This audit was conducted specifically for the Circle CCTP V2 integration review. The scope is limited to the smart contracts listed in Section 2.2 and does not cover off-chain components, frontend applications, or server-side infrastructure.

Binenet only issues this report based on the facts that have occurred or existed before the issue of this report, and assumes corresponding responsibilities for it. For the facts that occurred or existed after the issuance, we cannot judge the security status of the smart contract, and we will not be responsible for it.

This report does not include external contract calls beyond the CCTP interfaces, new types of attacks that may appear in the future, or contract upgrades or tampered codes. With the development of the project, smart contracts may add new modules, new external contract calls, etc.

The documents and materials provided to us by the information provider as of the date of this report. Binenet assumes that there is no missing, tampered, deleted or concealed information provided. If the information provided is missing, tampered, deleted, concealed or reflected inconsistent with the actual situation, Binenet shall not be liable for any losses and adverse effects resulting therefrom.

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12 About Binenet

Founded in June 2023, Binenet is a dedicated and pure blockchain security company, focusing on accurate, efficient and intelligent blockchain threat detection and response. Committed to providing users with professional products and dedicated services in the field of blockchain security. Business functions cover penetration testing, code auditing, emergency response, on-chain data monitoring, AML anti-money laundering, etc., covering all aspects of blockchain ecosystem security.

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